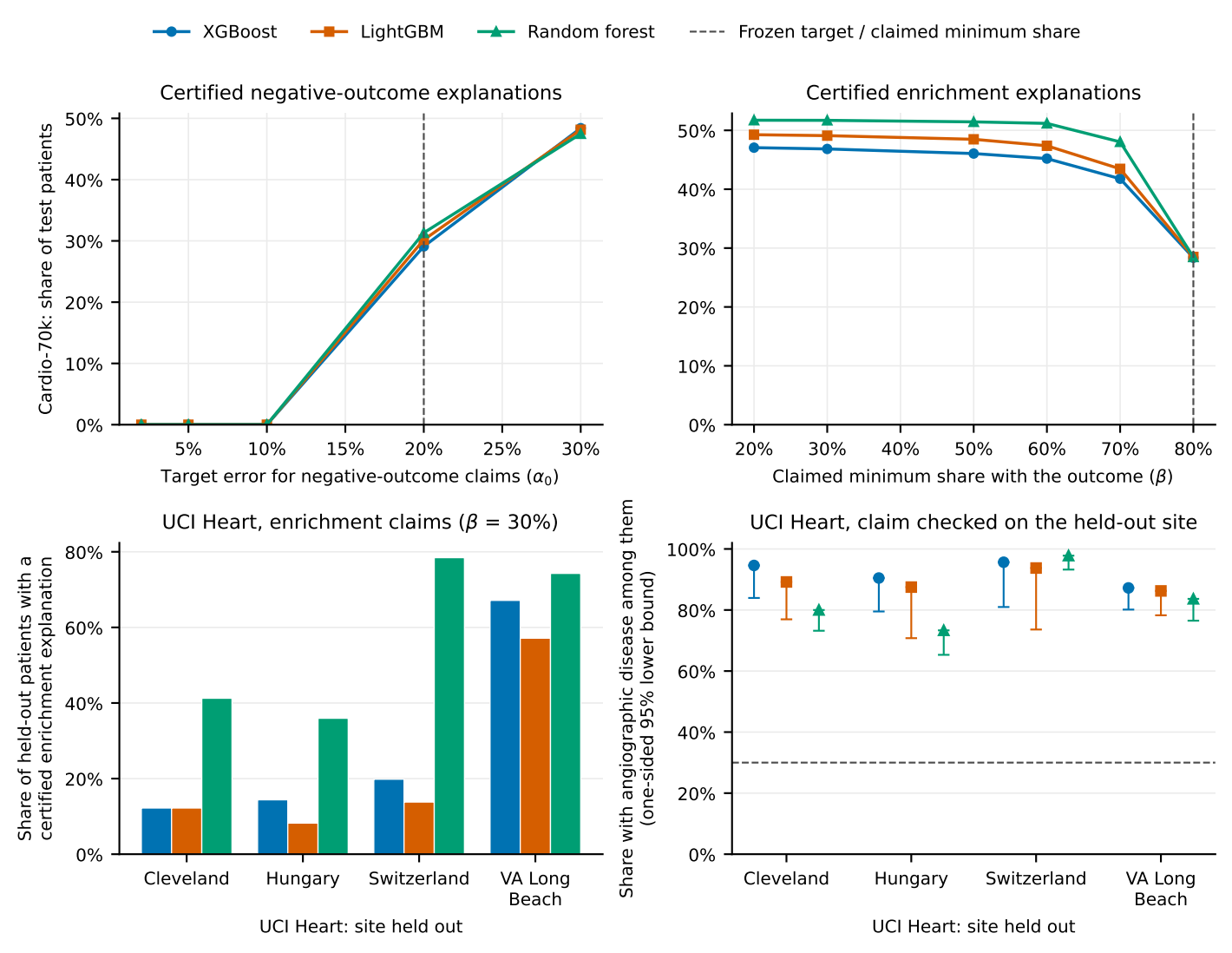
XGBoost
LightGBM
Random forest
Frozen target / claimed minimum share
Certified negative-outcome explanations
Cardio-70k: share of test patients
0%
10%
20%
30%
40%
50%
5%
10%
15%
20%
25%
30%
Target error for negative-outcome claims ($\alpha_0$)
Certified enrichment explanations
20%
30%
40%
50%
60%
70%
80%
Claimed minimum share with the outcome ($\beta$)
UCI Heart, enrichment claims ($\beta$ = 30%)
Share of held-out patients with a certified enrichment explanation
Cleveland
Hungary
Switzerland
VA Long Beach
UCI Heart: site held out
UCI Heart, claim checked on the held-out site
Share with angiographic disease among them (one-sided 95% lower bound)
100%
UCI Heart: site held out